Effect of Movement Strategy and Staffing Level on Simulated Contact Patterns

AMR-Hub agent-based model: contact analysis report

Abstract

This report compares four agent movement configurations in the AMR-Hub agent-based model and examines how simulated contact patterns scale with the number of healthcare workers (HCWs). Thirty seeded replicates were run for each combination of movement strategy (fixed speed, deadline-paced), movement stochasticity (on, off) and HCW count (2, 4, 6, 8, 16, 24 and 32), giving 840 runs on a synthetic scenario. Contact volume grows far faster than staffing: a 16-fold increase in HCWs (2 to 32) increased contact observations by about 63-79 times, driven by HCW-HCW pairs. Deadline-paced movement produced fewer contact observations than fixed-speed movement at every HCW count, but the relative reduction shrank from roughly 26-28% at 2 HCWs to about 10-17% at 32 HCWs, and it raised the HCW-HCW share of contacts. The results describe a small, synthetic, over-subscribed layout and must not be generalised to real hospitals.

1. Introduction

Contact between people is the pathway through which antimicrobial-resistant organisms spread in the model, so the way agents move determines the contact network on which transmission acts. Two questions motivate this analysis:

1. How does the choice of movement strategy change the amount and structure of contact?
2. How do contact metrics scale as the number of HCWs increases?

No transmission or infection dynamics are analysed here; the report concerns contact only.

2. Methods

2.1 Simulation set-up

All runs use the synthetic scenario in `python-code/tests/inputs/`: a single building layout, 4 patients and a simulation window of 06:55-19:05 with a 1-second timestep. One contact is recorded when a pair of agents is within the contact-distance threshold (0.1 metres) at a recorded timestep; pairs located in different known rooms are excluded. Episodes are maximal consecutive runs of contact for a pair, and durations are converted to seconds using the timestep.

2.2 Movement strategies

Strategy	Description
Fixed	Agents move at the configured movement speed.
Deadline-paced	Speed is the remaining distance divided by the time to the task's scheduled start.

Each strategy was run with movement stochasticity (standard deviation of the variability of angular deviations) on (configured value 0.5) and off (0), giving four cases.

2.3 Staffing levels

The 2-HCW condition is the original scenario. Larger conditions add synthetic HCWs produced by `contact_analysis/generate_extra_hcws.py`; each performs workstation work, patient rounds on the existing patients, and occasional WC and restaurant visits. The same 4 patients are used in every condition. The sweep was run with `contact_analysis/run_hcw_sweep.py`.

2.4 Replication and statistics

Each case and HCW count was replicated 30 times with fixed seeds. Reported values are means across replicates with 95% Student-t confidence intervals. Metrics are computed by `plot_contact_summary.py` and `plot_contact_types.py` and aggregated by `plot_hcw_sweep.py`, which also writes `python-code/contact_analysis/results/hcw_sweep_summary.csv`.

2.5 Metrics

- **Contact observations**: pair-timestep contacts per run. With a 1-second timestep this equals total pair-contact time in seconds.
- **Number of contact episodes** per run.
- **Mean and longest episode length** (seconds).
- **HCW-HCW share**: the percentage of contact observations that are between two HCWs (mean of per-run percentages).

3. Results

3.1 Scaling with number of HCWs

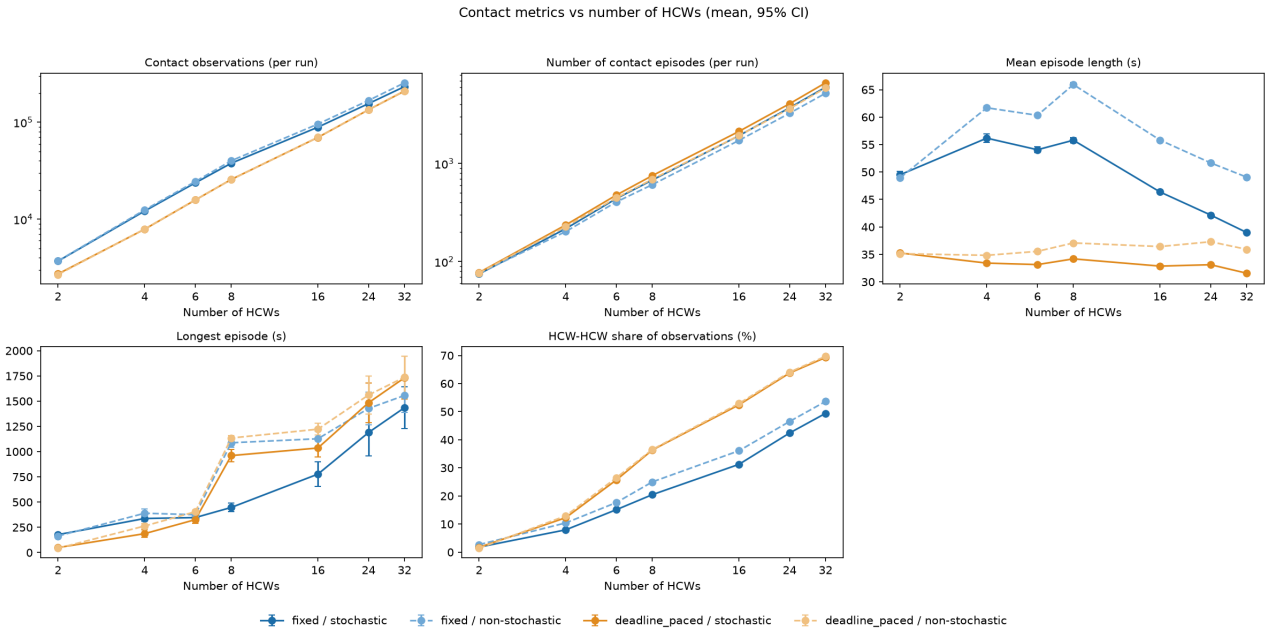


Figure 1. Contact metrics against HCW count for each strategy and stochasticity case.

Table 1 gives the mean contact observations and episodes per run at selected HCW counts.

HCWs	Fixed / on	Fixed / off	Deadline / on	Deadline / off
2	3,700	3,721	2,727	2,690
4	12,155	12,483	7,892	7,930
8	37,825	40,141	25,729	25,649
16	89,039	95,927	69,804	70,035
32	233,534	255,268	209,797	211,303

Table 1. Mean contact observations per run.

HCWs	Fixed / on	Fixed / off	Deadline / on	Deadline / off
2	74.8	76.0	77.4	76.6
4	216.6	202.3	236.3	227.7
8	678.2	608.9	752.7	691.6
16	1,920.6	1,718.6	2,124.0	1,922.5
32	5,990.8	5,202.6	6,644.9	5,882.7

Table 2. Mean number of contact episodes per run.

Table 3 gives the HCW-HCW share of contact observations.

HCWs	Fixed / on	Fixed / off	Deadline / on	Deadline / off
2	1.8%	2.7%	1.7%	1.4%
6	15.1%	17.7%	25.7%	26.5%
8	20.5%	25.0%	36.4%	36.5%
16	31.3%	36.1%	52.4%	52.9%
24	42.4%	46.5%	63.7%	64.1%
32	49.4%	53.6%	69.3%	69.7%

Table 3. HCW-HCW share of contact observations.

Key observations:

- **Superlinear growth.** For fixed speed with stochasticity on, contact observations rose from about 3.7 thousand (2 HCWs) to 234 thousand (32 HCWs), a factor of about 63 for a 16-fold increase in staff. This is consistent with the number of HCW pairs growing roughly quadratically, together with more HCW-patient overlap.
- **Episodes grow more slowly than contact time.** Episodes increased by about 80-fold (74.8 to 5,991) for fixed/on while contact observations increased by about 63-fold, but mean episode length fell at high HCW counts (fixed/on: 56 s at 4 HCWs to 39 s at 32 HCWs), so contact is increasingly fragmented into more, shorter episodes.
- **Longest episodes lengthen.** Longest episodes exceed 1,000 s in every case at 16 or more HCWs (for example fixed/on rises from 447 s at 8 HCWs to 1,190 s at 24 HCWs), consistent with sustained co-location in shared rooms.

3.2 Comparison of movement strategies

- **Contact volume.** Deadline-paced movement produced fewer contact observations than fixed speed at every HCW count. The reduction was about 26-28% at 2 HCWs, about 32-37% at 4-8 HCWs, and about 10% (stochastic on) to 17% (off) at 32 HCWs, so the strategy effect shrinks as the layout becomes crowded.
- **Episode count and length.** Deadline-paced movement gave more but shorter episodes (mean about 32-37 s at every HCW count) than fixed speed (about 39-66 s).
- **Pair-type composition.** The strategies differ in who contacts whom. At 32 HCWs, deadline-paced movement gave about 64 thousand HCW-patient observations against about 118 thousand for fixed speed (about 46% lower), but about 145-147 thousand HCW-HCW observations against 115-137 thousand. The HCW-HCW share is therefore consistently higher under deadline-paced movement (Table 3).
- **Stochasticity.** Switching stochasticity on or off changed results much less than changing the strategy. Non-stochastic fixed runs gave somewhat more contact and longer episodes; with deadline-paced movement the on/off difference in contact observations was small at 4 or more HCWs (about 1%).

3.3 Detailed small-population results

For the original scenario (2 HCWs, 4 patients) contact was almost entirely HCW-patient (97.3-98.6%); patient-patient contact did not occur (See Table 3). With 6 HCWs the HCW-HCW share rose to 15-27%. Figures broken down by pair type are in `contact-counts-by-type.png`, `contact-duration-by-type.png`, `contact-episodes-by-type.png` and `contact-summary.png` (2 HCWs), with the corresponding sets in `six-hcw/` and `thirty-two-hcw/`; see `contact-movement-strategy-report.md` for the full per-case tables.

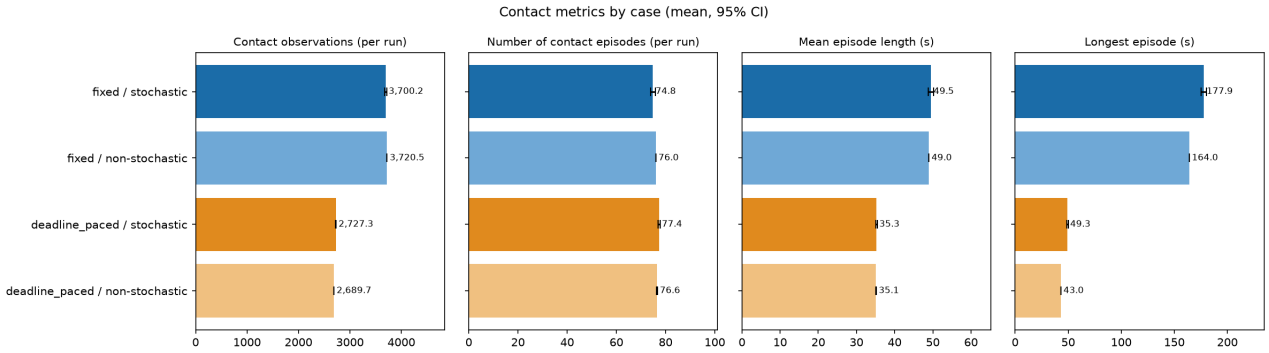


Figure 2. Contact metrics by case with 95% confidence intervals, 2 HCWs.

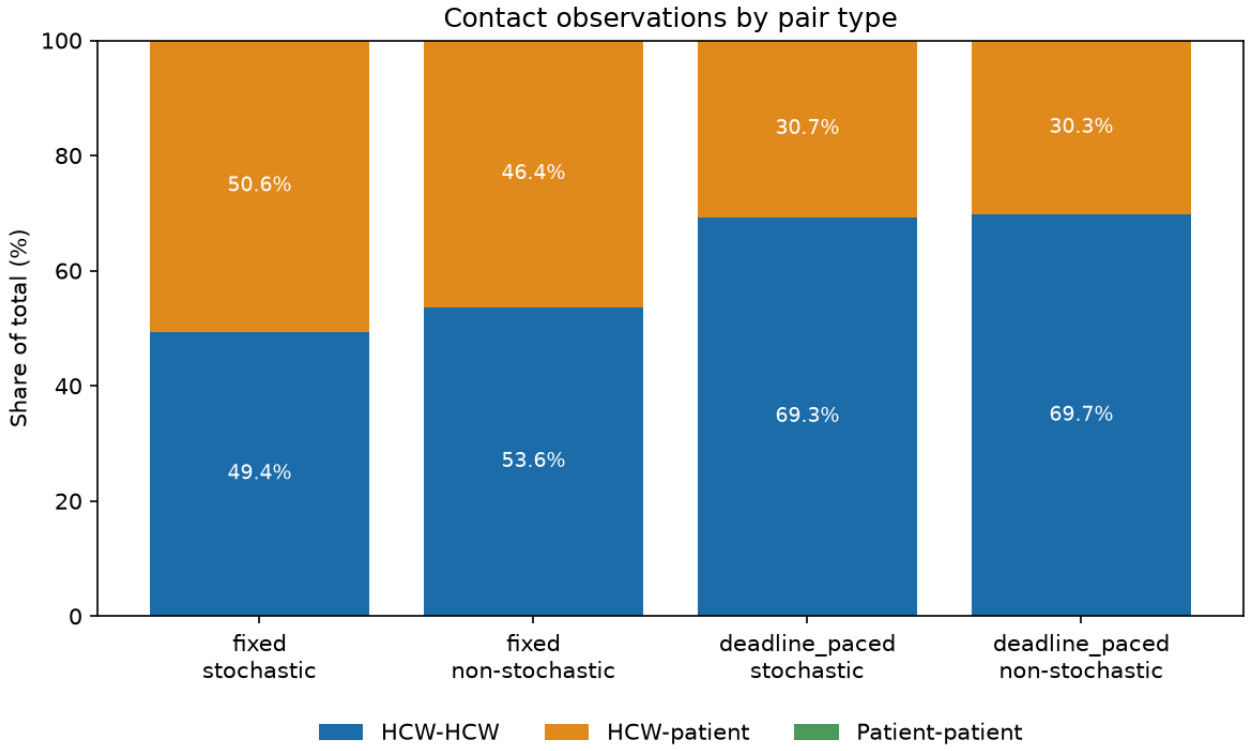


Figure 3. Share of contact observations by pair type, 32 HCWs.

3.4 Task completion

In the original 2-HCW runs every case completed all scheduled WC and restaurant occupancy tasks (8 WC and 8 restaurant visits per run, 100% of 16 tasks). Completion was not re-checked at higher HCW counts.

4. Discussion

Staffing level is the dominant driver of contact in this model: moving from 2 to 32 HCWs changes contact by almost two orders of magnitude, far more than the choice of movement strategy. Movement strategy nonetheless matters, mainly through composition rather than volume. Deadline-paced agents spend less time close to patients but more time close to colleagues, which would change which transmission routes (staff-to-staff versus staff-to-patient) dominate if infection dynamics were added.

The following explanations are hypotheses and have not been tested: the lengthening of the longest episodes and the shrinking strategy effect at high HCW counts probably reflect crowding in the small Staff Room and the shared WC and Restaurant, since with 32 HCWs there are only 3 workstations, one toilet and 6 chairs. Deadline-paced agents arrive just in time and so may cluster at shared resources, while fixed-speed agents arrive early and wait elsewhere.

5. Limitations

- The scenario is synthetic, with a single layout and only 4 patients; results should not be generalised to real hospital populations or workflows.
- Extra HCWs follow a generated schedule built for the scenario layout, and large counts over-subscribe shared resources (extra HCWs fall back to random points in the Staff Room when workstations are exhausted). High-HCW results show crowding effects rather than realistic staffing.
- Only contact is measured; no infection or transmission is modelled in these analyses.
- Confidence intervals reflect seed-to-seed variation only, not uncertainty in parameters such as the contact-distance threshold or movement stochasticity.
- Intermediate HCW counts (4, 8, 16, 24) are shown in Figure 1 and the tables above but do not have the per-pair-type figures of the 2, 6 and 32 HCW runs.

6. Reproducibility

Run from `python-code/` (see `contact_analysis/README.md`):

```
uv run python contact_analysis/run_hcw_sweep.py --hcw-counts 2 4 6 8 16 24 32 --replicates
30
uv run python contact_analysis/plot_hcw_sweep.py
```

Raw per-run metrics are in `python-code/contact_analysis/results/contact_pair_types_<N>hcw.csv` and the aggregated summary is in `hcw_sweep_summary.csv`. All inputs are synthetic.

7. Conclusions

1. Contact grows superlinearly with the number of HCWs, driven by HCW-HCW pairs.

2. Deadline-paced movement reduces total contact relative to fixed speed, but by a diminishing margin as staffing increases.
3. Deadline-paced movement shifts contact from HCW-patient to HCW-HCW pairs.
4. Movement stochasticity has a smaller effect than strategy or staffing.
5. Conclusions hold for the synthetic, over-subscribed scenario only.